9.3 Telescoping Sums Worksheet — Solutions

Telescoping Sums

Method: (1) Use partial fractions (or log rules) to write each term as a difference $u_k = b_k - b_{k+m}$. (2) Write out s_n and cancel the inner terms to get a closed form. (3) The sum of the series is $\lim_{n\to+\infty} s_n$; if the limit does not exist, the series diverges.

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1. $\displaystyle\sum_{k=1}^{\infty} \frac{1}{k^2+3k+2} = \frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \cdots + \frac{1}{k^2+3k+2} + \cdots$

Partial fractions. $k^2+3k+2 = (k+1)(k+2)$, and

$$\frac{1}{(k+1)(k+2)} = \frac{1}{k+1} - \frac{1}{k+2}.$$

Partial sum.

$$s_n = \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \cdots + \left(\frac{1}{n+1} - \frac{1}{n+2}\right) = \frac{1}{2} - \frac{1}{n+2}.$$

Limit. $\displaystyle\lim_{n\to+\infty} s_n = \lim_{n\to+\infty}\left(\frac{1}{2} - \frac{1}{n+2}\right) = \frac{1}{2}.$

Answer: $\boxed{\text{converges, sum} = \frac{1}{2}}$

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2. $\displaystyle\sum_{k=1}^{\infty} \frac{1}{(2k-1)(2k+1)} = \frac{1}{1\cdot 3} + \frac{1}{3\cdot 5} + \frac{1}{5\cdot 7} + \cdots$

Partial fractions.

$$\frac{1}{(2k-1)(2k+1)} = \frac{1}{2}\left(\frac{1}{2k-1} - \frac{1}{2k+1}\right).$$

Partial sum.

$$s_n = \frac{1}{2}\left[\left(1 - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{5}\right) + \cdots + \left(\frac{1}{2n-1} - \frac{1}{2n+1}\right)\right] = \frac{1}{2}\left(1 - \frac{1}{2n+1}\right).$$

Limit. $\displaystyle\lim_{n\to+\infty} s_n = \frac{1}{2}(1-0) = \frac{1}{2}.$

Answer: $\boxed{\text{converges, sum} = \frac{1}{2}}$

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3. $\displaystyle\sum_{k=1}^{\infty} \ln\left(\frac{k}{k+1}\right) = \ln\frac{1}{2} + \ln\frac{2}{3} + \ln\frac{3}{4} + \cdots$

Log rule. $\ln\left(\dfrac{k}{k+1}\right) = \ln k - \ln(k+1)$.

Partial sum.

$$s_n = (\ln 1 - \ln 2) + (\ln 2 - \ln 3) + \cdots + (\ln n - \ln(n+1)) = \ln 1 - \ln(n+1) = -\ln(n+1).$$

Limit. $\lim\limits_{n\to+\infty} s_n = \lim\limits_{n\to+\infty} (-\ln(n+1)) = -\infty$, so the limit does not exist.

Answer: $\boxed{\text{diverges}}$

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4. $\displaystyle\sum_{k=1}^{\infty} \frac{2}{k(k+2)} = \frac{2}{1\cdot 3} + \frac{2}{2\cdot 4} + \frac{2}{3\cdot 5} + \cdots$

Partial fractions.

$$\frac{2}{k(k+2)} = \frac{1}{k} - \frac{1}{k+2}.$$

Partial sum. Now terms cancel *two apart*, so two positive terms survive at the start and two negative terms at the end:

$$\begin{aligned} s_n &= \left(1 - \frac{1}{3}\right) + \left(\frac{1}{2} - \frac{1}{4}\right) + \left(\frac{1}{3} - \frac{1}{5}\right) + \cdots + \left(\frac{1}{n-1} - \frac{1}{n+1}\right) + \left(\frac{1}{n} - \frac{1}{n+2}\right) \\ &= 1 + \frac{1}{2} - \frac{1}{n+1} - \frac{1}{n+2}. \end{aligned}$$

Limit. $\lim\limits_{n\to+\infty} s_n = 1 + \dfrac{1}{2} - 0 - 0 = \dfrac{3}{2}$.

Answer: $\boxed{\text{converges, sum} = \dfrac{3}{2}}$

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5. $\displaystyle\sum_{k=1}^{\infty} \frac{4}{k^2 + 4k + 3} = \frac{4}{2\cdot 4} + \frac{4}{3\cdot 5} + \frac{4}{4\cdot 6} + \cdots$

Partial fractions. $k^2 + 4k + 3 = (k+1)(k+3)$, and

$$\frac{4}{(k+1)(k+3)} = \frac{2}{k+1} - \frac{2}{k+3}.$$

Partial sum. Terms cancel two apart:

$$\begin{aligned} s_n &= 2\left[\left(\frac{1}{2} - \frac{1}{4}\right) + \left(\frac{1}{3} - \frac{1}{5}\right) + \left(\frac{1}{4} - \frac{1}{6}\right) + \cdots + \left(\frac{1}{n+1} - \frac{1}{n+3}\right)\right] \\ &= 2\left(\frac{1}{2} + \frac{1}{3} - \frac{1}{n+2} - \frac{1}{n+3}\right). \end{aligned}$$

Limit. $\lim\limits_{n\to+\infty} s_n = 2\left(\dfrac{1}{2} + \dfrac{1}{3}\right) = 2\cdot\dfrac{5}{6} = \dfrac{5}{3}$.

Answer: $\boxed{\text{converges, sum} = \frac{5}{3}}$

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6. $\dfrac{1}{4\cdot 5}+\dfrac{1}{5\cdot 6}+\dfrac{1}{6\cdot 7}+\cdots+\dfrac{1}{(k+3)(k+4)}+\cdots$

Sigma form. This is $\displaystyle\sum_{k=1}^{\infty}\frac{1}{(k+3)(k+4)}$ (check: $k=1$ gives $\frac{1}{4\cdot 5}$).

Partial fractions.

$$\frac{1}{(k+3)(k+4)}=\frac{1}{k+3}-\frac{1}{k+4}.$$

Partial sum.

$$s_n=\left(\frac{1}{4}-\frac{1}{5}\right)+\left(\frac{1}{5}-\frac{1}{6}\right)+\cdots+\left(\frac{1}{n+3}-\frac{1}{n+4}\right)=\frac{1}{4}-\frac{1}{n+4}.$$

Limit. $\lim_{n\to+\infty} s_n=\dfrac{1}{4}$.

Answer: $\boxed{\text{converges, sum} = \frac{1}{4}}$

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7. $\dfrac{1}{1\cdot 4}+\dfrac{1}{4\cdot 7}+\dfrac{1}{7\cdot 10}+\cdots+\dfrac{1}{(3k-2)(3k+1)}+\cdots$

Sigma form. This is $\displaystyle\sum_{k=1}^{\infty}\frac{1}{(3k-2)(3k+1)}$ (check: $k=1$ gives $\frac{1}{1\cdot 4}$).

Partial fractions. The factors differ by 3, so

$$\frac{1}{(3k-2)(3k+1)}=\frac{1}{3}\left(\frac{1}{3k-2}-\frac{1}{3k+1}\right).$$

Partial sum.

$$s_n=\frac{1}{3}\left[\left(1-\frac{1}{4}\right)+\left(\frac{1}{4}-\frac{1}{7}\right)+\cdots+\left(\frac{1}{3n-2}-\frac{1}{3n+1}\right)\right]=\frac{1}{3}\left(1-\frac{1}{3n+1}\right).$$

Limit. $\lim_{n\to+\infty} s_n=\dfrac{1}{3}(1-0)=\dfrac{1}{3}$.

Answer: $\boxed{\text{converges, sum} = \frac{1}{3}}$